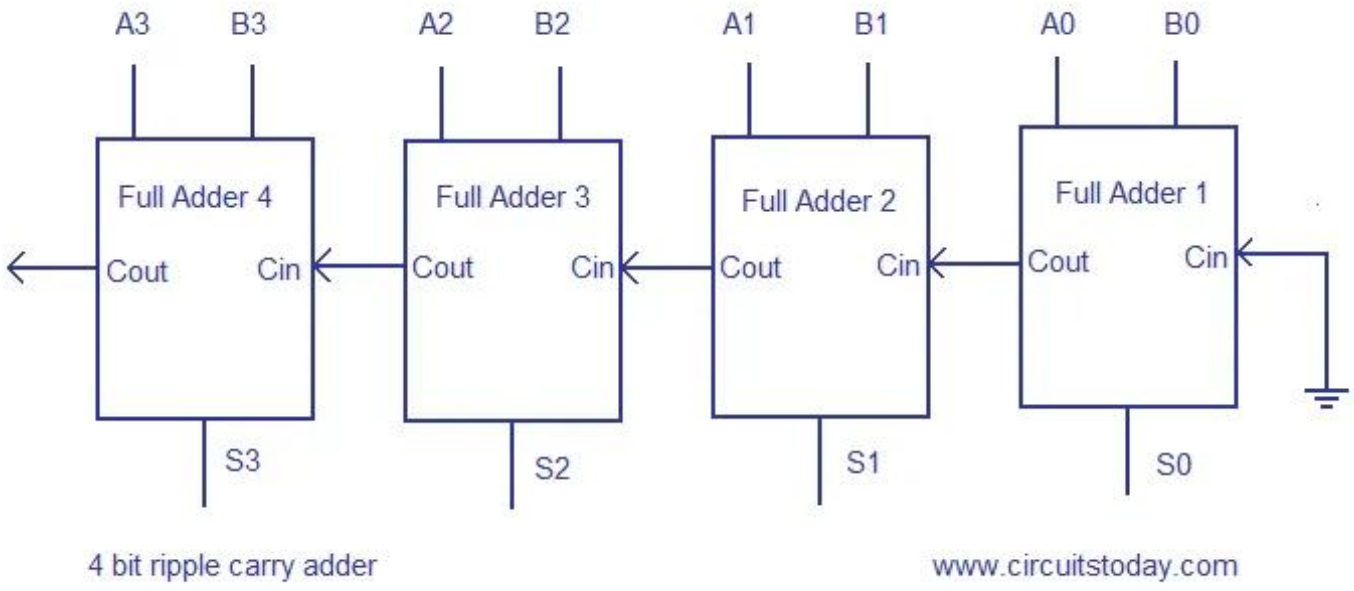
Program 1: Design and Implementation of a Full Adder Using VHDL

```
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;

entity full_adder is
    Port (
        A    : in  STD_LOGIC;
        B    : in  STD_LOGIC;
        Cin  : in  STD_LOGIC;
        Sum  : out STD_LOGIC;
        Cout : out STD_LOGIC
    );
end full_adder;

architecture Behavioral of full_adder is
begin
    Sum  <= A XOR B XOR Cin;   -- Sum is 1 for odd number of 1s
    Cout <= (A AND B) OR (Cin AND (A XOR B));  -- Carry when at least
                                                  two inputs are 1

end Behavioral;
```

Truth Table

A	B	Cin	Sum	Cout
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

Figure 1: Block Diagram of 4-Bit Ripple Carry Adder Using Four Full Adders



Program 2: Design and Implementation of a 4-Bit Ripple Carry Adder Using VHDL

```
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;

entity rca_4bit is
    Port (
        A   : in  STD_LOGIC_VECTOR(3 downto 0);
        B   : in  STD_LOGIC_VECTOR(3 downto 0);
        Cin : in  STD_LOGIC;
        Sum : out STD_LOGIC_VECTOR(3 downto 0);
        Cout : out STD_LOGIC
    );
end rca_4bit;

architecture Structural of rca_4bit is

    component full_adder
        Port (
            A   : in  STD_LOGIC;
            B   : in  STD_LOGIC;
            Cin : in  STD_LOGIC;
            Sum : out STD_LOGIC;
            Cout : out STD_LOGIC
        );
    end component;

    signal C : STD_LOGIC_VECTOR(3 downto 0); -- internal carry chain

begin
    -- four full adders chained: carry-out of each feeds carry-in of next

    FA0: full_adder port map(A(0), B(0), Cin, Sum(0), C(0));
    FA1: full_adder port map(A(1), B(1), C(0), Sum(1), C(1));
    FA2: full_adder port map(A(2), B(2), C(1), Sum(2), C(2));
    FA3: full_adder port map(A(3), B(3), C(2), Sum(3), Cout);

end Structural;
```

Program 3: Simulation and Verification of a 4-Bit Ripple Carry Adder Using Testbench

```
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;

entity tb_rca_4bit is      -- testbench, no ports
end tb_rca_4bit;

architecture Test of tb_rca_4bit is
      signal A, B   : STD_LOGIC_VECTOR(3 downto 0);
      signal Cin    : STD_LOGIC;
      signal Sum    : STD_LOGIC_VECTOR(3 downto 0);
      signal Cout   : STD_LOGIC;
begin
      -- device under test
      DUT: entity work.rca_4bit
           port map (
              A    => A,
              B    => B,
              Cin  => Cin,
              Sum  => Sum,
              Cout => Cout
           );
      -- apply test inputs
      process
      begin
           A <= "0101"; B <= "0011"; Cin <= '0';     -- 5 + 3 = 8
           wait for 10 ns;
           A <= "1111"; B <= "0001"; Cin <= '0';     -- 15 + 1 = 16 (overflow)
           wait for 10 ns;
           A <= "1010"; B <= "0101"; Cin <= '0';     -- 10 + 5 = 15
           wait for 10 ns;
           A <= "1111"; B <= "1111"; Cin <= '0';     -- 15 + 15 = 30 (overflow)
           wait for 10 ns;
           A <= "1010"; B <= "0011"; Cin <= '1';     -- 10 + 3 + 1 = 14
           wait for 10 ns;
           wait;
      end process;
end Test;
```

Figure 2: Simulation Waveform of 4-Bit Ripple Carry Adder in ModelSim